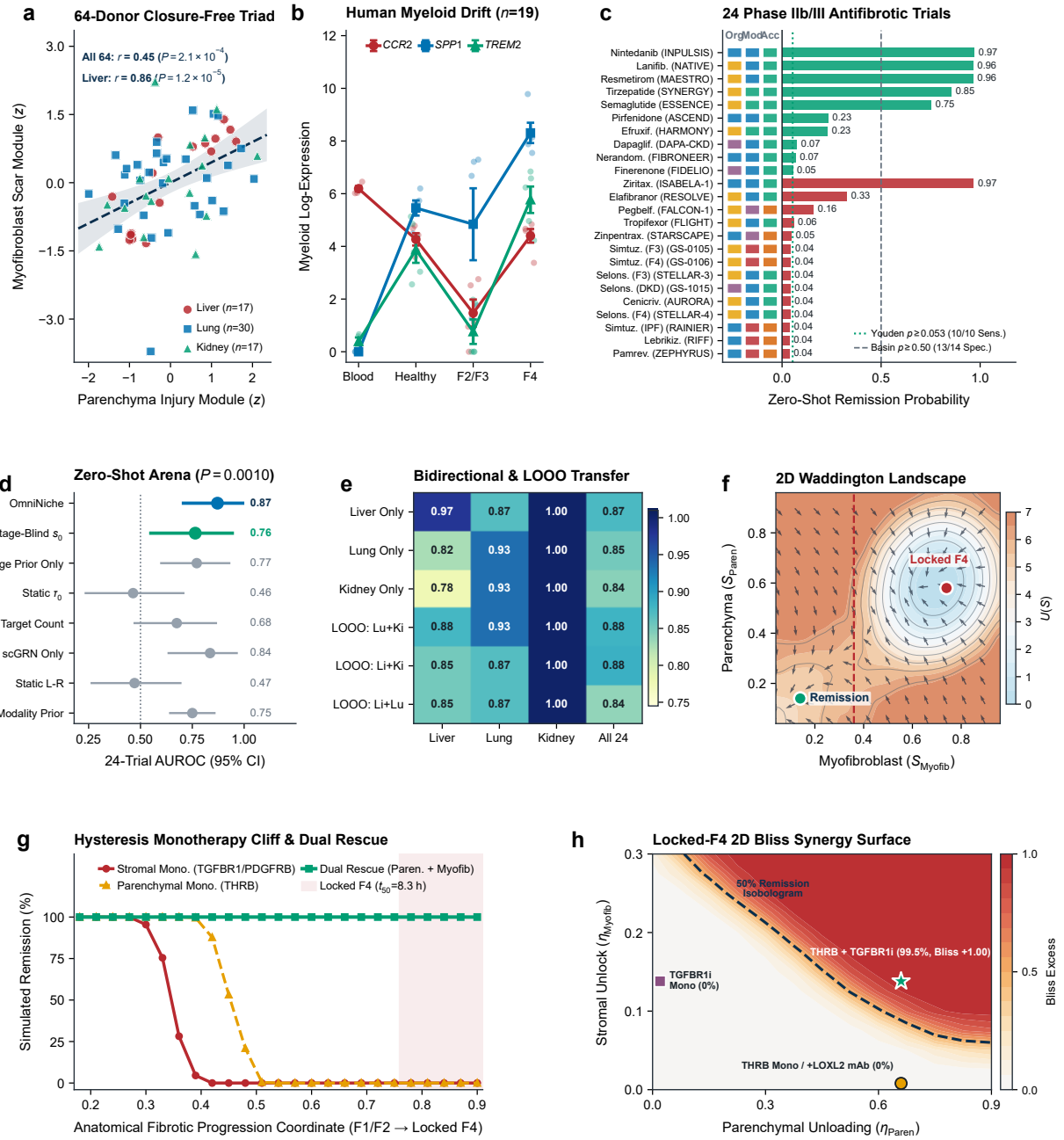

a
64-Donor Closure-Free Triad
All 64: r = 0.45 (P = 2.1 × 10−4)
Liver: r = 0.86 (P = 1.2 × 10−5)
Liver (n=17)
Lung (n=30)
Kidney (n=17)
Myofibroblast Scar Module (z)
Parenchyma Injury Module (z)
b
Human Myeloid Drift (n=19)
CCR2
SPP1
TREM2
Myeloid Log-Expression
Blood
Healthy
F2/F3
F4
c
24 Phase IIb/III Antifibrotic Trials
Org
Mod
Acc
Nintedanib (INPULSIS)
Lanifib. (NATIVE)
Resmetirom (MAESTRO)
Tirzepatide (SYNERGY)
Semaglutide (ESSENCE)
Pirfenidone (ASCEND)
Efruxif. (HARMONY)
Dapaglif. (DAPA-CKD)
Nerandom. (FIBRONEER)
Finerenone (FIDELIO)
Ziritax. (ISABELA-1)
Elafibranor (RESOLVE)
Pegbelf. (FALCON-1)
Tropifexor (FLIGHT)
Zinpentrax. (STARSCAPE)
Simtuz. (F3) (GS-0105)
Simtuz. (F4) (GS-0106)
Selons. (F3) (STELLAR-3)
Selons. (DKD) (GS-1015)
Cenicriv. (AURORA)
Selons. (F4) (STELLAR-4)
Simtuz. (IPF) (RAINIER)
Lebrikiz. (RIFF)
Pamrev. (ZEPHYRUS)
Youden p ≥ 0.053 (10/10 Sens.)
Basin p ≥ 0.50 (13/14 Spec.)
Zero-Shot Remission Probability
d
Zero-Shot Arena (P = 0.0010)
OmniNiche
Stage-Blind s0
Stage Prior Only
Static τ0
Target Count
scGRN Only
Static L-R
Modality Prior
24-Trial AUROC (95% CI)
e
Bidirectional & LOOO Transfer
Liver Only
Lung Only
Kidney Only
LOOO: Lu+Ki
LOOO: Li+Ki
LOOO: Li+Lu
Liver
Lung
Kidney
All 24
f
2D Waddington Landscape
Locked F4
Remission
Parenchyma (SParen)
Myofibroblast (SMyofib)
U(S)
g
Hysteresis Monotherapy Cliff & Dual Rescue
Stromal Mono. (TGFBR1/PDGFRB)
Dual Rescue (Paren. + Myofib)
Parenchymal Mono. (THRB)
Locked F4 (t50=8.3 h)
Simulated Remission (%)
Anatomical Fibrotic Progression Coordinate (F1/F2 → Locked F4)
h
Locked-F4 2D Bliss Synergy Surface
50% Remission Isobologram
THRB + TGFBR1i (99.5%, Bliss +1.00)
TGFBR1i Mono (0%)
THRB Mono / +LOXL2 mAb (0%)
Stromal Unlock (ηMyofib)
Parenchymal Unloading (ηParen)
Bliss Excess